

Cumulative Final Examination

Due: Wednesday, December 16 at 8:00 AM **Value: 150 points** **Format: Two-hour in-class final examination**

Purpose

Integrate structure, energy, information, communication, division, and experimental design across the semester.

Your task

- Section A (30): Answer six 5-point prompts: hydrophobic effect in folding; membrane selective permeability; vesicle targeting specificity; enzyme saturation; DNA proofreading; and checkpoint arrest. Each answer must name a structure and a mechanism.
- Section B1 (10): A mitochondrial inner-membrane protein knockout lowers oxygen consumption, ATP, and membrane potential but raises NADH. Build a causal sequence connecting all four observations.
- Section B2 (10): Predict whether an uncoupler restores oxygen consumption, ATP, both, or neither in the knockout and justify.
- Section B3 (10): Identify a positive control, loading control, and localization control for confirming the knockout.
- Section B4 (5): Name one result that would contradict your proposed location of the defect.
- Section C (30): A growth-factor receptor becomes ligand-independent. Trace expected changes through kinase signaling, transcription, nutrient uptake, ATP demand, G1/S control, and apoptosis. Identify one negative-feedback response and one therapeutic vulnerability.
- Section D (30): Protein Q falls after Drug A while its mRNA is unchanged. Design experiments distinguishing reduced translation from increased protein degradation. Include untreated and vehicle controls, a translation-rate measure, a degradation-rate measure, predicted outcomes, and one limitation.
- Section E (25): Write a structured synthesis explaining how membranes enable energy conversion and information processing. Use at least four specific systems chosen from mitochondria, chloroplasts, action potentials, receptor signaling, vesicle trafficking, and nuclear transport; identify the shared principle and one important difference.

Submit

- Completed final-examination booklet and approved two-sided reference sheet.

Important note

One instructor-issued two-sided reference sheet is allowed. Students may annotate it by hand. Basic calculators are permitted.

Grading criteria

Criterion	Pts	Full-credit evidence
Section A	30	Accurate mechanisms with relevant cellular locations.
Section B	35	Reads all panels and builds a supported explanation.
Section C	30	Connects molecular change to multiple cellular outcomes.
Section D	30	Feasible design, controls, predicted results, and limitations.
Section E	25	Coherent synthesis using four specific systems.